EDA(Exploratory Data Analysis)

Objective

The objective of this analysis is to explore employee data and identify key factors contributing to employee attrition.

Using SQL for data exploration and analysis, the goal is to uncover patterns related to demographics, job roles, compensation, and work environment that influence employee turnover, and provide insights to support data-driven decision-making.

The objective of this EDA was to profile a workforce of **1,470 employees** and identify high-risk attrition drivers. Initial findings revealed a significant **data skew in tenure and job levels**, necessitating a shift from raw counts to **normalized attrition rates** for accurate business insights.

Dataset Overview

The dataset consists of employee-level information used to analyze attrition patterns within the organization.

- Total Records: 1470 employees
- Total Features: 35 columns
- Target Variable: Attrition (indicates whether an employee has left the company)

The dataset includes variables related to employee demographics (age, gender), job-related factors (department, job role, job level), compensation (salary), and work environment factors (overtime, job satisfaction, work-life balance).

Data Cleaning & Quality Checks

Before performing the analysis, the dataset was examined to ensure data quality and reliability while preserving the integrity of the original raw data.

- Data Type Validation: Each column's data type was reviewed to ensure appropriate classification (numerical, categorical, ordinal) for accurate analysis.
- Duplicate Check: No duplicate records were identified in the dataset.
- Missing Values: No significant missing values were observed in key columns.
- Categorical Consistency: Categorical variables such as Attrition, Overtime, Department, and Job Role were reviewed to ensure consistent formatting and labeling.
- Ordinal Data Validation: Columns with ordinal values such as Job Level, Job Satisfaction, and Work-Life Balance were checked to ensure they fall within valid and expected ranges.

These steps ensured that the data used for analysis was accurate, consistent, and reliable.

Data Profiling & Distribution (Univariate Analysis)

To understand the "shape" of the organization, I performed a distribution audit across key demographics:

- **Gender:** The workforce is approximately **60% Male and 40% Female**.
- **Age & Tenure:** The average age is **36.9 years** and most employees fall within the 18–45 age range, indicating a relatively young workforce with a heavy skew in tenure toward the (**0–5) year bracket** (Avg: 7.0 years).
- **Income Levels:** Monthly income ranges from **$1,009 to $19,999**, with a mean of **$6,503**. The workforce is skewed toward lower earners. 749 employees (over 50%) fall into the "Low" income level, while only 195 are in the "High" level.
- **Organizational Structure:** Sales and Research & Development represent the largest departments, with **Laboratory Technicians and Sales Executives** being the most common roles.
- **Job Levels:** Most employees are at Level 1 (543) or Level 2 (534). There are very few senior leaders, with only 69 employees at Level 5.

Attrition Drivers (Bivariate & Correlation Analysis)

By correlating employee attributes with the **16.12% overall attrition rate**, several "High-Risk" triggers were identified:

A. The "Income-Risk" Correlation

- **The Overtime Correlation:** A critical positive correlation exists between Overtime and Attrition. Employees working Overtime exhibit an attrition rate of 30.5%, compared to 10.4% for those who do not—a nearly 3x multiplier.
- **Income Inverse Correlation:** Attrition count is highest at the Low Income ($<5,000) level. As Monthly Income increases, the probability of attrition decreases. Employees who left the company earned an average of **$4,787**, nearly **30% less** than the $6,832 average of those who stayed.
- **Job Level Variance:** Level 1 positions show the highest attrition count and rate, with stability increasing significantly at Level 2 and above.
- **Age Variance**: Employees who leave are younger on average (33.6 years) than those who stay (37.5 years).
- **Income Gap**: A sharp disparity exists; those who exit earn significantly less ($4,787 avg) compared to those who remain ($6,832 avg).
- **Tenure Variance:** Attrition is more frequent among employees with lower company experience (5.1 years avg) compared to retained staff (7.3 years avg).

B. Impact of Workplace Satisfaction & Balance

Using normalized rates to account for group size differences, the data revealed:

- **Work-Life Balance:** Employees rating their balance as '1' (Poor) showed a staggering **31.25% attrition rate**, double the company average.
- **Environment Satisfaction:** Poor environment ratings resulted in a **25.35% attrition rate**.
- **Job Satisfaction:** A low rating ('1') correlated with a **22.84% attrition rate**, highlighting a clear link between engagement and retention.

The data confirms a non-random distribution of turnover. Attrition is statistically clustered among younger, lower-paid employees in Level 1 roles who work Overtime and report low satisfaction scores.

Business Analysis

The Macro View (Baseline Metric)

- **Overall Attrition Rate: 16.12%**
- **The Gender Audit:** Attrition is statistically balanced across the total workforce (**Male: 17% | Female: 15%**).
- **The HR Exception:** A minor deviation exists in **Human Resources**, where **Females** exhibit a higher exit rate, suggesting a unique functional/cultural gap.

Workforce Segmentation (The "Who")

- **Generational Risk:** Attrition is highest among **Young Employees (18–30) at 25.91%**.
- **The Onboarding Gap:** Early tenure (**0–2 years**) shows a peak exit rate of **29.82%**.
- **Job Level Volatility: Entry-level (Level 1)** positions are the most unstable at **26.34%**.
- 👉 **Insight:** The organization faces a "leaky bucket" problem with early-career talent, suggesting a gap in long-term value propositions for new hires.

Operational Hotspots (The "Where")

- **Departmental Outliers:** Attrition is not uniform; it is concentrated in **Sales (39.76%)** and **Laboratory Technicians (23.94%)**.

- **The Overtime Multiplier:** In a bivariate analysis, **Sales Representatives working Overtime** reached a critical attrition rate of **66.6%**.
- 👉 **Insight:** Operational pressure (Overtime) is the primary "trigger" that turns a high-pressure role into an unsustainable one.

Diagnostic Drivers (The "Why")

- **The Overtime Correlation:** Employees working Overtime are **3x more likely to exit** (30.53%) than their peers (10.44%).
- **The Promotion–Salary Paradox:** Data reveals that employees in the **0–2 year promotion bracket** still exhibit high attrition.
    - **Root Cause:** Average salary for this group remains below retention thresholds. A title change without a competitive market adjustment is insufficient for retention.
- **Career Stagnation:** Stagnation becomes a flight risk after **6+ years** without a promotion, regardless of previous loyalty.
- **Sentiment Drivers:** Low **Work-Life Balance (31.25%)**, **Environment Satisfaction (25.35%)**, and **Job Satisfaction** scores serve as leading indicators for future turnover.

The "Perfect Storm" (Key Finding)

Through advanced SQL filtering and segment naming, I identified the most critical retention risk:

- **High-Risk Profile:** Employees with **Low Income + Overtime + 6+ Year Promotion Gap**.
- **Result: 70% Attrition Rate.**
- 👉 **Strategic Takeaway:** This segment represents a predictable failure point. Addressing even one factor (e.g., eliminating overtime for this group) could significantly lower the probability of exit.

Strategic Recommendations

- **Target the $5,000 Threshold:** Prioritize salary adjustments for Level 1 staff in **Sales and Lab** roles. Data shows that employees under this monthly income bracket are the primary flight risks.
- **Operational Overtime Caps:** Implement strict "Overtime Thresholds" for employees in their first **0–2 years**. The current 30.5% attrition rate for overtime workers proves the workload is burning out the new talent pipeline.

- **The "Promotion Plus" Policy:** Stop "Dry Promotions." Ensure every title change for early-tenure staff comes with a market-aligned salary bump to move them out of the high-risk income segment.
- **HR Cultural Audit:** Conduct a qualitative "Stay Interview" program specifically for **Females in HR**. Since this is a functional outlier not driven by the same pay/overtime issues as Sales, it requires a cultural intervention.
- **Stagnation Circuit Breaker:** Flag any employee reaching **6+ years without a promotion**. Automated career development reviews at year 5 can preempt the "Loyalty Exit" identified in the data.